Nathan Tsao

nathantsao.com | | | | US Citizen

Machine learning research engineer with two years of academic research and internship experience in applied machine learning and autonomous systems. Seeking a full-time machine learning engineer position in the U.S..

Education

University of Texas at Austin — Aug 2023 – May 2025
Masters of Science: Mechanical Engineering — GPA: 3.74
Thesis: Neural Port-Hamiltonian Differential Algebraic Equations

University of Illinois Urbana Champaign — Aug 2019 – May 2022
Bachelors of Science: Mechanical Engineering — GPA: 3.87

Work Experience

NASA Ames Research Center — June 2025 – Aug 2025
Autonomous Aircraft Operations Research Intern — Mountain View, CA

- Researching how traffic following behavior in dense airspaces affects aircraft performance metrics.
- Developing GPU-accelerated multi-modal traffic simulation of the Bay Area to guide infrastructure development of Unmanned Aerial Vehicles.

Berkeley Lights — June 2022 – Sep 2022
Hardware Engineering Intern — Berkeley, CA

- Developed mechanical and electrical design of temperature calibration sensor for the OptoSelect chip.
- Automated data acquisition and calibration pipeline using Python scripts.

Research Experience

Autonomous Systems Group: UT Austin — Dec 2023 – May 2025
Graduate Research Assistant — Austin, TX

- Developed a scalable electrical circuit modeling framework using compositional machine learning models.
- Created low-energy and fast-inference human activity recognition algorithm for batteryless sensors.

Hybrid Robotics Group: UC Berkeley — May 2022 – Jan 2023
Visiting Research Assistant — Berkeley, CA

- Led team to design the hardware and RL controller design of an actuated tail for quadruped robots.
- Created inexpensive 3D-printed quasi-direct-drive actuators with custom FOC motor controller.

RoboDesign Lab: UIUC — Jan 2022 – May 2022
Undergraduate Research Assistant — Urbana, IL

- Prototyped a low-cost force-sensing humanoid robot foot using elastomers and Hall sensors.

Preprints

Cyrus Neary*, Nathan Tsao*, and Ufuk Topcu. Neural Port-Hamiltonian Differential Algebraic Equations for Compositional Learning of Electrical Networks. *Under review at CDC 2025.*

Geffen Cooper*, Nathan Tsao*, Filippos Fotiadis, Ufuk Topcu, Radu Marculescu. Learning from Sparse and Asynchronous Data Streams for Batteryless Sensors. *Under review at NeurIPS 2025.*

Teaching Experience

ASE 370C Feedback Control Systems: UT Austin — Jan 2025 – May 2025
Graduate Teaching Assistant — Austin, TX

ME 314D Dynamics: UT Austin — Sep 2023 – Dec 2023
Graduate Teaching Assistant — Austin, TX

Honors and Awards

Dr. J. Parker Lamb Endowed Presidential Fellowship: UT Austin — 2023

Nominated by the Walker Department of Mechanical Engineering.

Highest Honors: UIUC — 2022

Awarded to students with at least a 3.8 Illinois GPA and continual commitment to service and education.

Dean's List: UIUC — 2019-22

Awarded to undergraduate students in the top 20 percent of their college class.

Outreach

Del Valle High School Visit — Mar 2025

Presented machine learning academic research to engineering and robotics high school students. — Austin, TX

UT Austin Girls Day — Feb 2024

Guided K-8th graders to program remote-controlled cars in Python. — Austin, TX

Skills

Skills: Deep Learning | Generative AI | Linux
Programming Languages: Python | Dart
Frameworks: PyTorch | Jax | Docker | Stable-Baselines3 | Git
Languages: English | Mandarin
Relevant Coursework: Reinforcement Learning | Statistical Machine Learning | Theoretical Statistics | Convex Optimization